Nonexistence of binary $[\![16, 5, 5]\!]$ and $[\![17, 3, 6]\!]$ stabilizer codes

Brandon Li
Independent researcher, Great Neck, New York
brandon.li.gn@gmail.com

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Abstract

We prove that binary stabilizer codes with parameters $[\![16, 5, 5]\!]$ and $[\![17, 3, 6]\!]$ do not exist, including degenerate codes. Together with known constructions, these results determine the optimal minimum distances $d_{\max}(16, 5) = 4$ and $d_{\max}(17, 3) = 5$. Both proofs use one method. We prove a necessary-and-sufficient extension criterion that expresses commutation and minimum distance through a linear map between quotient spaces, and a relative-distance version covering arbitrary degeneracy. We derive kernel and hull identities, a coset-counting obstruction, and a redundancy bound for a specified balanced local stabilizer. To apply the method, we first use weight-enumerator identities to restrict the stabilizer on a small set of coordinates. We then classify the possible shortened centralizers and exclude their extensions to the full coordinate set. The sixteen-qubit proof excludes extensions of 3,957 classified additive codes. For seventeen qubits, a forced weight-eight word and a quadratic-form argument reduce the proof to five local stabilizer types on eight coordinates. Complete classifications and exhaustive extension searches exclude all five types.

Keywords: quantum error correction; stabilizer codes; additive codes; coset distance; symplectic geometry; computer-assisted proof.

1 Introduction

An $[\![n, k, d]\!]$ quantum code uses n physical qubits to store k logical qubits. Its distance d is the smallest number of qubits on which a Pauli operator can act nontrivially on the encoded information while commuting with every stabilizer. Distance measures the weight of the smallest undetected logical error. A code of distance d corrects arbitrary errors on up to $\lfloor (d-1)/2 \rfloor$ qubits. Both distance five and distance six correct two-qubit errors; distance six also detects errors of weight five.

For fixed numbers of physical and logical qubits, determining the largest possible minimum distance is a basic problem in quantum coding theory. Constructions give lower bounds, while weight-enumerator identities and other general bounds give upper bounds. When these differ by one, the remaining existence question may depend on structure that an ordinary weight enumerator does not record.

Let $d_{\max}(n, k)$ denote the largest minimum distance of a binary stabilizer code encoding k logical qubits into n physical qubits. We prove the following two nonexistence results.

Theorem 1.1. *There is no binary stabilizer code with parameters* $[\![16, 5, d \geq 5]\!]$ *or* $[\![17, 3, d \geq 6]\!]$. *Hence*

$$d_{\max}(16, 5) = 4, \qquad d_{\max}(17, 3) = 5.$$

The conclusions hold for both pure and degenerate stabilizer codes.

The matching constructions are listed in Grassl's tables [3]. Their binary generators are in `verification/data/`, and their distances are checked by complete centralizer enumeration. The $[\![16, 5, 4]\!]$ construction extends a $[\![14, 5, 4]\!]$ code by two coordinates; the $[\![17, 3, 5]\!]$ construction similarly extends a $[\![15, 3, 5]\!]$ code. The two exclusions close the displayed distance intervals 4–5 and 5–6, respectively.

The proof begins by forcing a stabilizer word whose support leaves few enough complementary coordinates for a complete classification. Let T be that support and R the entire stabilizer supported on T. To shorten the centralizer on T, retain only its words that are identity on every coordinate of T, then delete those coordinates. Call the resulting additive code C. Every centralizer word has a component on T and a component on its complement. Once the first component is fixed modulo R, the second belongs to a single coset of C. The possible codes can be described by linear maps between quotient spaces.

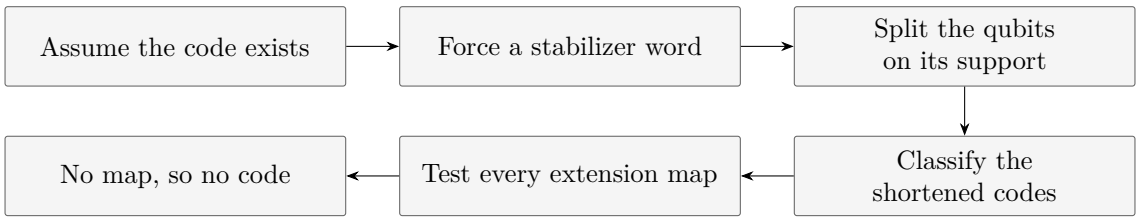


Figure 1: The structural reduction and the finite extension test.

A first reading can follow the definitions in Section 2 and the fourteen-qubit example in Section 4, where the counting inequality finishes the proof. Sections 5 and 6 apply the same setup to the two main targets; Section 7 explains why their finite lists are complete.

1.1 The extension criterion

A coset of a subspace consists of all vectors obtained by adding its members to one fixed vector. The quotient is the vector space of these cosets: it treats two vectors as equivalent when their difference lies in that subspace. Here this identifies local Pauli words that differ by a stabilizer supported on the selected coordinates.

Put $E = R_T^{\perp}/R$. Write $c_R(e)$ for the minimum weight of a representative of $e \in E$, and write $\delta(u)$ for the minimum weight of a coset u of C. In suitable coordinates, an extension is described by a binary matrix M satisfying

$$\Gamma = MJM^{\mathsf{T}}, \qquad c_R(e) + \delta(Me) \geq d \quad (e \neq 0). \tag{1}$$

Here J is the symplectic matrix on E, and Γ is the Gram matrix of a basis of $C^{\perp}$. The first equation is the commutation condition. The second states that the least weight in each corresponding pair of cosets is at least d.

Theorem 2.1 proves that these conditions are necessary and sufficient when $d(C) \geq d$ and every stabilizer word of weight below d lies in R. This includes pure codes and codes whose degeneracy is confined to the selected coordinates. Theorem 2.2 gives a relative-distance version without this restriction. In that version, the relevant coset minima also depend on M.

For example, suppose $R = \langle ZZZZ \rangle$ on four coordinates. Here I denotes the identity and Z a nonidentity Pauli letter. The local class $ZIII + R = \{ZIII, IZZZ\}$ has minimum weight one. If its image under M had coset minimum weight three, the two parts could be combined into a centralizer word of weight four outside R. In the setting of Theorem 2.1, this rules out distance five. The inequality in (1) makes this test for every local class at once.

Every nonzero element of $\ker M$ represents a logical operator supported entirely on T. The weights and symplectic products of these operators restrict the hull $C \cap C^{\perp}$ of the shortened code. A further counting condition compares cosets of C of small weight with classes in $E/\ker M$ of sufficiently large weight. These restrictions can exclude a shortened code before its extension matrices are enumerated.

1.2 Applications: the two nonexistence proofs

For sixteen qubits, exact weight-enumerator certificates and an integer congruence show that a hypothetical code is pure and has minimum stabilizer weight $m \in \{6, 7, 8, 9\}$. Shortening the centralizer on the support of such a word gives one of four additive-code classifications, containing 3,957 equivalence classes altogether. None admits a map satisfying the distance inequalities in (1). The commutation equation is not needed in this case: even this relaxation has no solution.

For seventeen qubits, purity alone leaves a less convenient collection of shortening cases. We instead prove that every hypothetical code has a stabilizer word of weight eight. A quadratic-form argument bounds the binary dimension of the shortened nine-coordinate code by six. It follows that there are just five possible local stabilizers R: one of rank one and four of rank two. The rank-one type has five admissible kernel cases, determined by the kernel dimension, restricted symplectic rank, and shortened-code hull. These restrictions reduce the remaining problem to complete classifications on nine coordinates and finite extension searches.

As a smaller application, Section 4 gives the fourteen-qubit argument in the same notation. There a unique weight-four stabilizer word forces $\ker M = 0$. Of the 37 possible shortened codes, 25 have the wrong hull dimension and the remaining 12 violate the coset-counting condition. The final contradiction requires no search over extension matrices.

1.3 Previous work

The symplectic representation of stabilizer codes and the MacWilliams and shadow identities are standard [1, 2]. Gundersen et al. [5] describe puncturing directly in symplectic and stabilizer matrix notation. Complementary symplectic projections are treated in [6]; related subspace constructions appear in the use of Goursat's lemma for subsystem codes [7] and in entanglement-assisted error correction [8]. Our extension criterion uses this linear algebra while retaining the minimum weights in the original coordinate blocks. Its role here is to derive the kernel and coset-counting restrictions and to specify the finite problems used in the two nonexistence proofs.

Classification by extension and the use of automorphism groups are also established methods [9]. We use the distance-five additive-code classifications of Grassl et al. [12], and give completeness checks for the additional classifications needed here. Earlier quantum nonexistence proofs combining structural reductions and finite searches include the $[\![13, 5, 4]\!]$ result of Bierbrauer et al. [10]. Related geometric methods for purity and local structure are developed in [11].

The semidefinite-programming bounds with rational certificates in [13], Table 1, leave code-space dimension $K = 32$ possible at $(n, d) = (16, 5)$, and $K = 8$ possible at $(17, 6)$: their displayed intervals are $32 \le K \le 42$ and $4 \le K \le 8$, respectively. Theorem 1.1 excludes the corresponding stabilizer codes by using their additional algebraic structure.

Singh [15] announced the $[\![14, 3, 5]\!]$ nonexistence result, including degenerate codes, on 10 August 2026. Section 4 gives a hull-and-coset argument as a worked application of the extension criterion.

Lai [16] develops refined weight-enumerator and structural tests. His Section VII-C1 also proves the nonexistence of $[\![14, 3, 5]\!]$ using the simplex mapping and binary shadow-coset counting. His Table II records the intermediate candidate counts; Section VII-D leaves 27 candidate distributions for $[\![16, 5, 5]\!]$ and seven for $[\![17, 3, 6]\!]$, with nonexistence unestablished for either parameter set. The present proofs exclude both parameter sets through complete shortened-code classifications and extension tests.

The computational premises, verification coverage, and data limits are described in Section 7 and Appendix C.

1.4 Organization

Section 2 proves the extension criterion. Section 3 derives the kernel, hull, and coset-counting bounds, including a redundancy bound when the local stabilizer is a full-support plane. Sections 4, 5, and 6 give the applications in increasing computational complexity. Section 7 describes classification completeness and the computational verification. The appendices specify the exact certificates, the extension algorithms, and the available data.

2 An extension criterion for stabilizer codes

2.1 Pauli space, distance, and degeneracy

All vector spaces are over $\mathbb{F}_2$. Write

$$V_n = \mathbb{F}_2^{2n} = \{(x \mid z) : x, z \in \mathbb{F}_2^n\}, \qquad \langle (x \mid z), (x' \mid z') \rangle = x \cdot z' + z \cdot x'.$$

At each coordinate, the pairs $(0,0), (1,0), (0,1), (1,1)$ represent I, X, Z, Y, respectively, with scalar phases omitted. Vector addition then represents multiplication of Pauli operators up to phase. The

symplectic product is zero for commuting operators and one for anticommuting operators. The support of a word is the set of coordinates where its letter is not I. The Pauli weight is $\mathrm{wt}(x \mid z) = |\{i : (x_i, z_i) \neq (0,0)\}|$. A Z-only word has letters I and Z, so its X-component vanishes. A stabilizer is represented by its binary subspace; Pauli phases are omitted. A subspace is isotropic if every pair of its words has symplectic product zero. A binary $[\![n, k]\!]$ stabilizer is an isotropic subspace $S \leq V_n$ of dimension $n-k$. Its centralizer is $D = S^\perp$, of dimension $n+k$. For a Pauli stabilizer this commuting space is also called its normalizer. For $k > 0$ its distance is

$$d(S) = \min\{\mathrm{wt}(w) : w \in S^\perp \setminus S\}.$$

The encoded states are the common eigenspace of independent commuting generators. The centralizer contains the Pauli operators commuting with all these generators; its elements outside S act nontrivially on the encoded information. Stabilizer words themselves act trivially there, so a low-weight stabilizer word is permitted. A code is pure to distance d when every nonzero word of its centralizer has weight at least d. For a code of distance at least d, degeneracy at that threshold means that S contains a nonzero word of weight below d. An additive code here is an $\mathbb{F}_2$-linear subspace of Pauli space; it need not be isotropic. For an additive code C, $d(C)$ means the minimum weight of a nonzero codeword, with $d(\{0\}) = +\infty$. The notation $(\ell, 2^c, d)_4$ means an additive code of length ℓ, binary dimension c, and minimum distance d over the four-letter Pauli alphabet.

Write A_j for the number of stabilizer words of weight j; their list is the weight enumerator. Two additive codes are physically equivalent if a permutation of coordinates and independent permutations of X, Y, Z at each coordinate carry one onto the other. These relabelings preserve weight and commutation.

Split the coordinates into T, of size m, and its complement, of size $f = n - m$. Write $V_n = V_T \oplus W$, with $W = V_f$. We view V_T and W as subspaces of V_n by padding with identity letters on the other coordinates. Write π_T and π_W for the two coordinate projections. A prefix is a word on T, and a suffix is a word on its complement.

Symbol	Meaning
n, k, d	Physical qubits, encoded qubits, and target distance
$S, D = S^\perp, A_j$	Stabilizer, centralizer, and stabilizer weight counts
T, m, f	Selected support, its size, and complementary size $n - m$
R, r	Stabilizer supported on T, and its dimension
$C, P = C^\perp$	Shortened centralizer, and its dual on the complement
E, a, b	Local quotient $R_T^\perp / R$, its dimension, and $\dim P$
M, τ, Γ	Extension map, its symplectic adjoint, and the Gram matrix on P
K, q, t	Kernel of M, its dimension, and its restricted form rank
N, h	Kernel of τ, and hull dimension $\dim(C \cap C^\perp)$
H, Q	Local centralizer $S^\perp \cap V_T$, and syndrome image $\mathrm{im}\, M$
σ, c_R, δ	Syndrome map, local coset minimum weight, and complementary coset minimum weight
$\varepsilon, \mathcal{K}_j^\ell$	Weight-parity function and quaternary Krawtchouk polynomial
μ	Half the local quotient dimension in the search appendix

2.2 The local quotient and the shortened code

The local data have different roles. The space $R = S \cap V_T$ records stabilizer words supported on T; the shortened centralizer $C = S^\perp \cap W$ records centralizer words supported on its complement. We now prescribe R and C and ask which full stabilizers can have these two spaces. Fix an isotropic

subspace $R \le V_T$ of dimension r, where $0 \le r \le m$, and define

$$E = R_T^{\perp}/R, \qquad a = \dim E = 2m - 2r.$$

Here $R_T^{\perp}$ is the orthogonal complement of R within V_T. An element of E is a coset $v + R$: two local operators are identified when their difference is a local stabilizer. This identification retains their possible logical action, while the function below retains their least physical weight. The quotient carries a nondegenerate symplectic form. Fix a symplectic basis with matrix $J = \left[\begin{smallmatrix} 0 & I \\ I & 0 \end{smallmatrix}\right]$. For $e \in E$, define

$$c_R(e) = \min_{v+R=e} \mathrm{wt}(v).$$

c_R is the minimum coset weight in the original coordinate block. An arbitrary symplectic change of basis need not preserve it.

The dimension required of C follows from projection and rank–nullity. For an actual extension, a vector on T is orthogonal to $\pi_T S^{\perp}$ if and only if it lies in $S \cap V_T = R$. $\pi_T S^{\perp} = R_T^{\perp}$, of dimension $2m - r$. Its projection kernel is C, so $\dim C = (n + k) - (2m - r)$. Fix an additive code $C \le W$ of binary dimension $n + k - 2m + r$, assumed to lie between zero and $2f$. Put

$$P = C_W^{\perp}, \qquad b = \dim P = n - k - r.$$

Orthogonal complements of C and P are taken inside W. Choose a basis $v_1, \ldots, v_b$ of P, set $\Gamma_{ij} = \langle v_i, v_j \rangle$, and identify W/C with $\mathbb{F}_2^b$ by the surjective syndrome map

$$\sigma(w)_j = \langle v_j, w \rangle, \qquad \delta(u) = \min\{\mathrm{wt}(w) : \sigma(w) = u\}.$$

Its kernel is C. In particular $\delta(0) = 0$, regardless of $d(C)$. A syndrome u is a label for one coset of C, and $\delta(u)$ is the smallest weight of any word with that label.

For a subspace U, its radical is $\mathrm{rad}(U) = \{u \in U : \langle u, v \rangle = 0 \text{ for every } v \in U\}$. For a code, this same intersection with its dual is called its hull. The nullspace of Γ represents $\mathrm{rad}(P) = P \cap P^{\perp} = C^{\perp} \cap C$. The binary nullity of Γ is $h = \dim(C \cap C^{\perp}) = \dim \mathrm{Hull}(C)$.

For reference, the four spaces used in the extension criterion are:

Space	Meaning	Binary dimension
R	Stabilizer words supported on T	r
E	Local operators commuting with R, modulo R	$a = 2m - 2r$
C	Centralizer shortened on T	$n + k - 2m + r$
$P = C^{\perp}$	Projections of stabilizer words onto the complement	$b = n - k - r$

2.3 Reconstructing the full stabilizer

The matrix M specifies which suffix coset accompanies each local class. Its entries alone do not guarantee a code: the two coordinate blocks must give commuting stabilizer words, and their combined weights must satisfy the distance requirement. These are the two conditions in the following theorem.

Theorem 2.1 (Stabilizer extension criterion)**.** *Assume $0 < k < n$ and $d(C) \ge d$, with R, C and their dimensions as above. There is a bijection between:*

(i) isotropic stabilizers S of dimension $n - k$ satisfying

$$S \cap V_T = R, \quad S^{\perp} \cap W = C, \quad \mathrm{wt}(v) \ge d \text{ for every } v \in S^{\perp} \setminus R;$$

(ii) matrices $M \in \mathbb{F}_2^{b\times a}$ satisfying

$$\Gamma = MJM^{\mathsf{T}}, \qquad c_R(e) + \delta(Me) \geq d \quad (0 \neq e \in E). \tag{2}$$

The stabilizer corresponding to M is $S_M = D_M^{\perp}$, where

$$D_M = \{(v, w) : v \in R_T^{\perp},\ \sigma(w) = M(v + R)\}. \tag{3}$$

The weight hypothesis in (i) is equivalent to distance at least d together with localization of every stabilizer word of weight below d inside R.

Proof. Recovering the attachment map. Let S satisfy (i), and put $D = S^{\perp}$. The annihilator of $\pi_T D$ inside V_T is $S \cap V_T = R$; hence $\pi_T D = R_T^{\perp}$. The kernel of $\pi_T|_D$ is C. For each $v \in R_T^{\perp}$, the corresponding words on the complement of T form a nonempty coset of C. Addition in D makes this coset assignment linear. Since $R \subseteq S \subseteq D$, it depends only on $e = v + R$, defining $M : E \to W/C$.

For $e \neq 0$, its prefix representatives and the members of its suffix coset can be chosen independently, by adding R and C. Their minimum total weight is $c_R(e) + \delta(Me)$. All lie outside R, proving the inequality. The suffix-only fiber gives $d(C) \geq d$.

Recovering the commutation condition. The duality between projection and shortening also gives $\pi_W S = P$. Indeed, a word on W annihilates $\pi_W S$ if and only if its extension by zeros belongs to $S^{\perp} \cap W = C$, so $(\pi_W S)_W^{\perp} = C$. The kernel of $\pi_W|_S$ is R, so S/R is the graph of a linear map $\tau : P \to E$. Write A for its a-by-b coordinate matrix. For $x = \sum c_j v_j$, this graph has prefix class Ac. Commutation with a centralizer word of prefix class e gives $e^{\mathsf{T}}JAc + c^{\mathsf{T}}Me = 0$ for every e, c. Thus $M = A^{\mathsf{T}}J$, equivalently $A = JM^{\mathsf{T}}$. Two graph words commute if and only if their local and complementary symplectic products cancel. Isotropy of the graph gives $\Gamma = A^{\mathsf{T}}JA = MJM^{\mathsf{T}}$. τ is the symplectic adjoint of M. Its kernel has a direct interpretation: $\tau(x) = 0$ means that a stabilizer projecting to x has local part in R, which can be removed by adding that local stabilizer. $\ker \tau = S \cap W$.

Constructing a stabilizer from the map. Conversely, let M satisfy (ii) and define D_M by (3). Its projection is $R_T^{\perp}$, its suffix-only kernel is C, and it contains R. Consequently $\dim D_M = 2m - r + \dim C = n + k$, so $\dim S_M = n - k$, and annihilator duality gives $S_M \cap V_T = R$. For a projected stabilizer word $x = \sum c_j v_j$, commutation with D_M forces its prefix class to be $JM^{\mathsf{T}}c$. The symplectic product of two such words is $c^{\mathsf{T}}(MJM^{\mathsf{T}} + \Gamma)c' = 0$. The local words of R commute with them because their prefixes lie in $R_T^{\perp}$. S_M is isotropic.

Every fiber of nonzero prefix class has weight at least d by (ii). In the zero-class fiber, the prefix lies in R and the suffix lies in C; a word outside R has a nonzero suffix, of weight at least d. This proves (i). The fibers determine D and therefore $S = D^{\perp}$ uniquely, establishing the bijection. □

2.4 A relaxation using only coset weights

The distance inequalities do not generally imply commutation. For example, take $n = 4, k = 1, d = 2$, $R = \langle ZZ \rangle$, and $C = \langle XX, ZZ \rangle$ on the other two sites. In the prefix basis Z_0, X_0X_1, the nonzero coset minimum weights are $1, 2, 2$. Choose independent nonzero syndrome columns. All weight inequalities hold, since the three nonzero cosets of C have distance one. But C is self-dual, so $\Gamma = 0$, whereas $MJM^{\mathsf{T}} \neq 0$. Excluding all weight-admissible maps still excludes genuine extensions. Finding one such map would not construct a code.

2.5 Extensions of degenerate codes

For an arbitrary degenerate code, C may contain short stabilizer words. The same graph construction still applies, with the distance taken relative to the reconstructed stabilizer.

Theorem 2.2 (Relative extension criterion)**.** *Keep R, C, E, P and their dimensions, but impose no lower bound on $d(C)$. Let M satisfy $\Gamma = MJM^{\mathsf{T}}$, and define $\tau_M(\sum c_j v_j) = JM^{\mathsf{T}}c$, and*

$$\delta_M^{\log}(e) = \min\{\mathrm{wt}(w) : \sigma(w) = Me,\ w \notin \tau_M^{-1}(e)\},$$

where $\tau_M^{-1}(e)$ is a subset of P, and the minimum of the empty set is $+\infty$. Then S_M has distance at least d if and only if

$$c_R(e) + \delta_M^{\log}(e) \geq d \quad \text{for every } e \in E,$$

including $e = 0$.

Proof. The graph and dimension arguments above do not require a distance assumption. A word $(v, w) \in D_M$, with $v+R = e$, is in S_M if and only if $w \in P$ and $\tau_M(w) = e$. Membership is independent of the representative of e, because $R \subseteq S_M$. The displayed minimum ranges over precisely the suffixes of logical words in that fiber. Prefix and suffix representatives vary independently. Minimizing their sum gives the assertion, fiber by fiber. □

At $e = 0$, the suffix minimum is the distance of C relative to $N = \ker \tau_M$: it is $\min\{\mathrm{wt}(w) : w \in C \backslash N\}$. The coset minimum depends on M, so the theorem does not supply a fixed precomputed table of coset minimum weights for arbitrary degeneracy. The searches below use Theorem 2.1 only after proving purity or localization.

3 Restrictions on extension kernels

We derive necessary conditions from the kernel and image of the extension map. Throughout this section, S is a stabilizer with $S \cap V_T = R$, $S^{\perp} \cap W = C$, and extension map M. Every nonzero kernel element represents a logical operator supported on T. There are two kernels to distinguish. The kernel $K = \ker M$ records logical operators supported on T. The kernel $N = \ker \tau$ records stabilizer words supported on the complementary coordinates: their projected local class is zero, and an element of R removes the remaining local representative. The hull identities below relate these two kinds of local information.

Proposition 3.1 (Kernel and hull identities)**.** *For an extension, put $K = \ker M$, $q = \dim K$, and let t be the rank of the symplectic form restricted to K. Then*

$$\operatorname{rank} \Gamma = a - 2q + t, \qquad h = b - a + 2q - t, \tag{4}$$

$$N := \ker \tau \subseteq \mathrm{rad}(P), \qquad \dim N = b - a + q. \tag{5}$$

Every nonzero $e \in K$ has $c_R(e) \geq d(S)$, even without purity. Moreover $q - t/2 \leq k$.

Proof. The adjoint map has image $K_E^{\perp}$, of dimension $a - q$. To see this directly, e annihilates $\operatorname{im} \tau$ if and only if $\langle e, \tau(x) \rangle = 0$ for every $x \in P$, or equivalently $Me = 0$. Nondegeneracy of E then gives $\operatorname{im} \tau = K_E^{\perp}$. The radical of its restricted form is $K \cap K^{\perp}$, of dimension $q - t$ by rank–nullity on the restricted form. Pullback along the surjection $P \to K^{\perp}$ leaves the form rank unchanged, giving

$a - q - (q - t)$. Subtracting from b gives (4). The map τ has rank $a - q$, giving (5); a vector in its kernel pairs trivially with all of P.

For $e \neq 0$ in K, a prefix representative with zero suffix lies in $S^{\perp}$. It cannot lie in S, because $S \cap V_T = R$. Its minimum weight is at least the logical distance. The space K embeds in the $2k$-dimensional nondegenerate logical space $S^{\perp}/S$. A subspace of dimension q and form rank t has an isotropic subspace of dimension $q - t/2$; that dimension is at most k. □

3.1 Counting cosets of small weight

Let $Q = \operatorname{im} M$. The syndrome subspace Q is naturally $N_W^{\perp}/C$: these are precisely the suffix classes annihilating $N = \ker \tau$. The map M induces a bijection $E/K \to Q$. For $\eta \in E/K$, put $\kappa(\eta) = \min_{e \in \eta} c_R(e)$. $\kappa(\eta)$ is the least weight available among all local classes that have the same suffix image. If that suffix image has small minimum weight, every one of its possible local partners must compensate with a sufficiently large weight. Counting these partners gives the next inequality.

Proposition 3.2 (Coset-counting inequality)**.** *In the localized setting of Theorem 2.1, for* $0 \leq j < d$*,*

$$|\{0 \neq u \in Q : \delta(u) \leq j\}| \leq |\{0 \neq \eta \in E/K : \kappa(\eta) \geq d - j\}|. \tag{6}$$

If every nonzero class in E has minimum weight below d, then

$$K = 0, \quad h = b - a, \quad N = \operatorname{rad}(P), \quad Q = \operatorname{im} \Gamma.$$

Proof. For a fixed η, all its members have the same syndrome image. Minimizing (2) over them gives $\kappa(\eta) + \delta(M\eta) \geq d$. Each nonzero syndrome class of minimum weight at most j therefore corresponds to a distinct nonzero coset in E/K whose minimum weight is at least $d - j$. This proves the inequality. The minimum-weight hypothesis and Proposition 3.1 imply $K = 0$. Then (4) and (5) give the remaining statements, with $\operatorname{im} \Gamma = (\ker \Gamma)^{\perp}$ in syndrome coordinates. □

Violation of (6) excludes an extension without enumerating the matrices M. The inequality alone is not sufficient for existence.

3.2 A binary bound for a rank-one local stabilizer

Proposition 3.3 (Kernel projection bound)**.** *Suppose $R = \langle Z^m \rangle$ and $d > \lfloor m/2 \rfloor$. The physical X-projection maps K injectively to an even binary linear code of length m, dimension q, and distance at least $\Delta = 2(d - \lfloor m/2 \rfloor)$. Consequently,*

$$m \geq \sum_{i=0}^{q-1} \left\lceil \frac{\Delta}{2^i} \right\rceil. \tag{7}$$

Proof. Adding Z^m leaves the X-mask unchanged, so this projection is well defined on E. Commutation with Z^m makes its weight w even. If z counts Z-letters off that mask, then

$$c_R(e) = w + \min(z, m - w - z) \leq \lfloor m/2 \rfloor + w/2.$$

A nonzero class represented by a Z-only word has minimum weight at most $\lfloor m/2 \rfloor$, so cannot lie in K. This proves injectivity, and $c_R(e) \geq d$ gives $w \geq \Delta$. The last inequality is the binary Griesmer

bound applied to that projected code. For completeness, choose a minimum-weight binary word of weight w. Puncturing its support leaves a code of dimension one less and distance at least $\lceil w/2 \rceil$: the two representatives of a nontrivial quotient class have weight sum equal to w plus twice their common suffix weight, and each has weight at least w. Only zero and the chosen word can lie wholly in that support. Induction gives the stated sum. □

The binary Griesmer bound applies to the projected kernel. For $m = 8, d = 6$, it gives $q \leq 4$. Section 6 sharpens this to three using parity and the logical symplectic dimension.

3.3 Balanced local planes and redundancy

When the local stabilizer is generated by two full-support words, the minimum-weight condition determines the symbol counts of every nonzero kernel class. This gives the following restriction.

Theorem 3.4 (Balanced-plane obstruction)**.** *Suppose $S \cap V_T = R = \langle X^m, Z^m \rangle$, where $m = 4\omega$, $\omega \geq 1$, and $d(S) \geq 3\omega$. Then $2^q \leq m$. If ω is even, K is isotropic. If ω is odd, then $q \leq 2$, with restricted form rank zero for $q \leq 1$ and two for $q = 2$. In particular,*

$$\begin{aligned} n - k &\geq 2m - 2 - \lfloor \log_2 m \rfloor, && \omega \text{ even}, \\ n - k &\geq 2m - 4, && \omega \text{ odd}. \end{aligned}$$

No purity assumption is needed.

Proof. For a word on T, let n_I, n_X, n_Y, n_Z be its four symbol counts. Addition of R permutes the four counts, so the minimum weight modulo R is $m - \max n_L \leq 3m/4$. A nonzero kernel class has minimum weight at least 3ω; hence every count is ω. All four words in that class have weight 3ω.

Let $\widetilde{K}$ be the preimage of K in $R_T^{\perp}$, and put $L = |\widetilde{K}| = 2^{q+2}$. Its weight enumerator is $1 + (L - 4)z^{3m/4} + 3z^m$. The second MacWilliams coefficient of its dual equals

$$\frac{6m^2}{L} - \frac{3m}{2}.$$

This coefficient counts weight-two words in the dual code. It is nonnegative, so $L \leq 4m$, or $2^q \leq m$.

The Pauli parity function $\varepsilon(v) = \mathrm{wt}(v) \bmod 2$ satisfies

$$\varepsilon(u + v) = \varepsilon(u) + \varepsilon(v) + \langle u, v \rangle. \tag{8}$$

If ω is even, all words of $\widetilde{K}$ are even, so $\widetilde{K}$, and hence K, is isotropic. If ω is odd, two distinct nonzero kernel classes pair to one, since the three representatives involved have odd weight. Three independent classes would contradict bilinearity: $\langle u, v + w \rangle = 1 + 1 = 0$, although the same rule requires one. $q \leq 2$, with the asserted ranks.

Here $a = 2m - 4$, $b = n - k - 2$. In the even case, $\operatorname{rank} M = a - q \leq b$ gives $n - k \geq 2m - 2 - q$. In the odd case, the smallest possible $\operatorname{rank} \Gamma = a - 2q + t$ is $a - 2$, so $b \geq a - 2$, giving the second bound. □

For example, the specified local plane on twelve coordinates at distance at least nine requires $n - k \geq 20$. The hypothesis fixes the entire stabilizer supported on those twelve coordinates; the conclusion applies to codes with that local structure.

3.4 Choosing a tractable support

Proposition 3.5 (Support selection)**.** *Let $B(f,d)$ be an upper bound on the binary dimension of an additive length-f, minimum-distance-d code. If the stabilizer of a pure $[\![n,k,d]\!]$ code contains a nonzero word with support T, where $|T|=m$, then $r=\dim(S\cap V_T)$ satisfies*

$$1\le r\le B(n-m,d)-n-k+2m.$$

Proof. The shortened centralizer has dimension $n+k-2m+r$ by Theorem 2.1's projection argument, and distance at least d by purity. Apply the additive dimension bound. □

The selected word need not have minimum stabilizer weight. This freedom is useful when an exact enumerator certificate forces a particular weight whose complement is already classifiable.

4 A supporting application: fourteen qubits

We apply the kernel and coset-counting conditions to fourteen qubits. The existence question is Research Problem 1 of Ball, Centelles, and Huber [4]. The parameter conclusion was previously announced by Singh [15]. Lai [16], Section VII-C1, also proves this nonexistence result using the simplex mapping and binary shadow-coset counting. We give a hull-and-coset proof as an application of the extension criterion. This example shows that the kernel conditions can exclude extensions without enumerating their matrices, even when the code is degenerate.

Proposition 4.1 (Fourteen-qubit local reduction)**.** *A hypothetical $[\![14,3,d\ge 5]\!]$ code has exactly one nonzero stabilizer word of weight at most four. That word has weight four.*

Proof. The ordinary enumerator/shadow identity in the repository gives

$$11685888A_1+1345536A_2+4681728A_3+989184A_4\le 3336192.$$

Integrality implies $A_1=A_3=0$, $A_2\le 2$, $A_4\le 3$. A further exact congruence gives $A_3+A_4\equiv 1 \pmod 2$, hence $A_4\in\{1,3\}$. It is obtained from the total row, low-weight MacWilliams rows, and signed shadow equalities with multipliers

$$(1,-2,-6,-17,-95;\ -20,-24,1108,246)$$

in the order total; low rows one through four; shadow equalities zero through three. Modulo 4096 only the coefficients of A_3,A_4 survive, both equal to 2048, and the right-hand side is 2048.

An exact split-coset certificate excludes a weight-two stabilizer word. For distinct commuting weight-four words, let ι be their support intersection size and u the number of different letters on that intersection. Then u is even and their sum has weight $8-2\iota+u$. All cases with $\iota\ge 2$ are excluded as follows:

ι	u	Obstruction
4	2	Their sum has weight two.
4	4	Certificate for $\langle XXXX, ZZZZ\rangle$.
3	0	Their sum has weight two.
3	2	Certificate for $\langle XXZZI, ZZZIZ\rangle$.
2	0	Certificate for $\langle ZZZZII, ZZIIZZ\rangle$.
2	2	Certificate for $\langle ZZZZII, XXIIXX\rangle$.

If $\iota = 4, u = 0$, the words are identical. Distinct weight-four words intersect in at most one site, with equal letters there. All such words may be made Z-only by one consistent local Clifford map.

If $A_4 = 3$, the three words are independent: the sum of two has weight six or eight, never four. Their three supports have pairwise intersection sizes zero or one. The intersection graph has zero, one, two, or three edges; when it has three, either the intersections are distinct or there is one common site. These give exactly five support geometries. Their union sizes are 12, 11, 10, 9, and 10, respectively. Exact rank-three coset certificates exclude each geometry, with the disjoint one using three four-site blocks and a two-site suffix. All include the already proved zeros and the assumption $A_4 = 3$. The data are in `verification/certificates/14/`, with the standalone checker `verification/check_certificates14.py`. $A_4 = 1$, proving the assertion. □

Normalize the unique short word to $p = Z^4$, let T be its support, and put $R = \langle p \rangle$. No other local stabilizer word is possible, since all words on T have weight at most four. The extension theorem applies with localized degeneracy, even though the code is not pure. Here

$$a = 6, \quad b = 10, \quad C \le V_{10}, \quad \dim C = 10, \quad d(C) \ge 5.$$

Every class in E has minimum weight at most four, forcing $K = 0$. Thus $\dim \mathrm{Hull}(C) = b - a = 4$ by Proposition 3.2.

The minimum weights of the 64 classes in E have the distribution

Minimum weight	0	1	2	3	4
Classes	1	4	27	24	8

To obtain it directly, group commuting words by their even X-mask. The zero mask contributes $1, 4, 3$ classes of minimum weight zero, one, and two. Each of the six weight-two masks contributes four classes of minimum weight two and four of minimum weight three. The full mask contributes eight classes of minimum weight four. With $J_C = \mathrm{Hull}(C)$, the coset-counting inequalities on $J_C^{\perp}/C$ become

$$\#\{u \ne 0 : \delta(u) \le 1\} \le 8, \quad \#\{u \ne 0 : \delta(u) \le 2\} \le 32, \quad \#\{u \ne 0 : \delta(u) \le 3\} \le 59.$$

There are 37 physical equivalence classes of additive $(10, 2^{10}, \ge 5)_4$ codes [12]. Their hull dimensions have multiplicities $0 : 7,\ 2 : 14,\ 4 : 12,\ 6 : 2,\ 8 : 2$. 25 classes fail the hull requirement. Every one of the remaining 12 violates at least one of the three coset-counting inequalities, as verified by enumerating physical suffix words and their cosets. No matrix-entry search is needed at this final step. This excludes $[\![14, 3, 5]\!]$; the known $[\![14, 3, 4]\!]$ construction gives $d_{\max}(14, 3) = 4$.

The sixteen-qubit $m = 6$ case uses the same 37 ten-coordinate codes, with a larger local quotient and different coset minimum weights.

5 The sixteen-qubit exclusion

Assume that $S \le V_{16}$ defines a $[\![16, 5, d \ge 5]\!]$ code. Then $|S| = 2^{11} = 2048$. The counts A_j include products of basis generators. The argument has three steps. First we prove purity and restrict the minimum stabilizer weight. Next we show that shortening on such a word leaves one of four complete code classifications. Finally we test the extension inequalities on every class in those lists.

5.1 Purity and minimum stabilizer weight

Proposition 5.1 (Exact low-weight reduction)**.** *Such a code is pure to distance five. Its minimum nonzero stabilizer weight m belongs to* $\{6, 7, 8, 9\}$*.*

Proof. The weight-enumerator equations count words of each weight without specifying the full code. The MacWilliams identities express centralizer counts in terms of stabilizer counts, and the shadow identities give additional nonnegative counts. A certificate is an integer combination of these necessary relations that gives a contradiction under a stated case assumption. Its check uses integer arithmetic alone; the general construction is given in Appendix A.

The six integer certificates in the repository exclude, in order:

(a) a stabilizer word of weight one, two, or three;
(b) a weight-four stabilizer word together with $A_4 \geq 2$;
(c) a weight-five stabilizer word under $A_1 = \cdots = A_4 = 0$;
(d) $A_1 = \cdots = A_9 = 0$.

For a proposed word of weight w, local Clifford operations normalize it to Z^w. The split system counts complete cosets of $\langle Z^w \rangle$, with distance equations through weight four, centralizer inclusion, and nonnegative shadow coefficients. Every actual code yields a feasible point. Appendix A defines the system and the exact arithmetic checks; the case assumptions and negative certificate values are tabulated there.

It remains to exclude $A_4 = 1$. Write $\mathcal{K}_j(w)$ for the quaternary Krawtchouk polynomial at length sixteen, and Sh_j for the shadow counts. Set

$$\Delta_0 = \sum_w A_w = 2048, \quad \Delta_j = \sum_w \mathcal{K}_j(w) A_w - 2048 A_j = 0 \ (1 \leq j \leq 4),$$

and $\Omega_j = \sum_w (-1)^w \mathcal{K}_j(w) A_w - 2048 \,\mathrm{Sh}_j = 0$. In the integer combination

$$-740\Delta_0 - 265\Delta_1 - 81\Delta_2 - 19\Delta_3 - 3\Delta_4 + 80\Omega_0 + 10\Omega_1 - 2\Omega_3,$$

all coefficients vanish modulo 4096, except those of A_1, A_2, A_3, A_4, each congruent to 2048. The right-hand side is $-740 \cdot 2048$, which is divisible by 4096. $A_1 + A_2 + A_3 + A_4$ is even. The first certificates give $A_1 = A_2 = A_3 = 0$, $A_4 \leq 1$, so $A_4 = 0$. Every centralizer word of weight below five must be a stabilizer; therefore purity follows. The last two certificate cases give $6 \leq m \leq 9$. □

5.2 The four shortened-code classifications

Choose a weight-m word p with support T, and normalize $p = Z^m$. We first need the *entire* local stabilizer.

Lemma 5.2. *For $m \in \{6, 7, 8, 9\}$, $S \cap V_T = \langle p \rangle$.*

Proof. If $\dim(S \cap V_T) \geq 2$, the shortened centralizer contains a subcode with parameters

$$(10, 2^{11}, \geq 5)_4, \quad (9, 2^9, \geq 5)_4, \quad \text{or} \quad (8, 2^7, \geq 5)_4$$

for $m = 6, 7, 8$, respectively. No additive codes with these parameters exist [12]; the repository also verifies their emptiness through the complete lengthening census. For $m = 9$, an independent local stabilizer word ζ and $\zeta + p$ must both have weight nine by minimality of m. ζ has X or Y at every site of T. It anticommutes with Z^9 on an odd number of sites, contradicting isotropy. □

The shorter code C has length $16 - m$, dimension $22 - 2m$, and distance at least five. The quotient W/C has dimension ten for every case. The complete classifications contain the following numbers of physical equivalence classes:

m	Shortened length	Binary dimension	Classes
6	10	10	37
7	9	8	2,298
8	8	6	1,579
9	7	4	43

The four class counts are the exact entries of Table 1 in Grassl et al. [12]. The repository reconstructs representatives and checks their completeness by a second physical graph encoding and orbit–stabilizer mass identities. The classification procedure is specified in Section 7.1.

5.3 The finite prefix-map problem

For $R = \langle Z^m \rangle$, use the symplectic basis

$$Z_0, \ldots, Z_{m-2}, \qquad X_i X_{m-1} \quad (0 \le i < m-1)$$

of E, with $Z_{m-1} = \sum_{i<m-1} Z_i$ modulo R. Each basis image is a ten-bit syndrome. Every actual code gives a linear map satisfying

$$c_R(e) + \delta(Me) \ge 5 \qquad (e \ne 0). \tag{9}$$

Only classes of minimum weight less than five constrain the search. Omitting $\Gamma = MJM^{\mathsf{T}}$ enlarges the feasible set, so nonexistence of a map in this larger set is sufficient.

The depth-first traversal assigns basis images and checks each newly determined subset sum. The forward-checking version maintains every remaining basis image's allowed domain. A value is removed only when a specific already determined subset would violate (9). Every word is tested: each $e \in E$ is a binary sum of the basis vectors, and its image is the corresponding sum of assigned syndromes. A Boolean satisfiability (SAT) encoding uses Boolean entries of M, XOR chains for its values, and a clause for every forbidden element/syndrome pair. Each forbidden pair is a clause requiring the entries of M to avoid that particular map value. The two encodings construct their coset labels separately.

Lemma 5.3 (Physical normalization for the weight-only search)**.** *If a map satisfying* (9) *exists, one exists with*

$$\xi_0 < \xi_1 < \cdots < \xi_{m-1}, \qquad \eta_i < \eta_i + \xi_i \quad (0 \le i < m-1),$$

where $\xi_i = M(Z_i)$, $\eta_i = M(X_i X_{m-1})$, *and labels use any fixed total order.*

Proof. The singleton and pair constraints force all ξ_i to be distinct and nonzero. A physical permutation of the m sites fixes Z^m and preserves coset minimum weights, so orders the singleton images. After this permutation, the map is re-expressed in the displayed standard basis; the dependent image is still $\xi_{m-1} = \sum_{i=0}^{m-2} \xi_i$. The phase operation $X \leftrightarrow Y$ on site $i < m-1$ fixes every Z_j, fixes the other complementary basis vectors, and replaces $X_i X_{m-1}$ by $X_i X_{m-1} + Z_i$. It exchanges η_i and $\eta_i + \xi_i$ independently for each i. These physical operations preserve all weight inequalities. □

Proposition 5.4 (Complete sixteen-qubit extension exclusion)**.** *None of the 3,957 shortened-code classes admits a map satisfying* (9)*.*

Computer-assisted proof. The recorded reports cover completed searches for every class. For $m \leq 8$, a symmetry-free depth-first search and a separately constructed SAT encoding both exclude every class. For $m = 9$, two depth-first implementations use Lemma 5.3, with different coset coordinates and search orders. Both exclude all 43 classes. A known pure $[\![16, 4, 5]\!]$ code is accepted by both encodings, with and without their symmetry restrictions. Appendix B specifies the finite predicate and completeness argument; Appendix C identifies the evidence and its checks. □

Proposition 5.1, Lemma 5.2, and Theorem 2.1 put every hypothetical code into one of these weight-admissible extension problems. Proposition 5.4 excludes them all. This proves the first assertion of Theorem 1.1.

6 The seventeen-qubit exclusion

Assume that $S \leq V_{17}$ is a $[\![17, 3, d \geq 6]\!]$ stabilizer, so $\dim S = 14$ and $\dim S^{\perp} = 20$. The proof first forces a weight-eight stabilizer word. Shortening on its support leaves nine coordinates, and the dimension bound below permits one rank-one local stabilizer type and four rank-two types. The rank-one type then splits into five kernel cases indexed by (q, t, h). We use *type* for the local stabilizer up to physical equivalence, *kernel case* for these three invariants, and *weight profile* for the pair (A_6, A_7). A search part fixes a shortened code, kernels, and the remaining identification or syndrome data.

Local type or kernel case	Exclusion	Shortened-code classes
Rank two: $(6, 6, 8)$	Counting leaves 48; extension tests	56
Rank two: $(6, 8, 8)$	Counting leaves 52; extension tests for each weight profile	56
Rank two: $(7, 7, 8)$	Counting leaves 52; extension tests	56
Rank two: $(8, 8, 8)$	Hull restrictions and fixed-kernel extension tests	56
Rank one: $(q, t, h) = (1, 0, 1)$	Fixed-kernel extension tests	26,752
Rank one: $(2, 2, 1)$	Fixed-kernel extension tests	26,752
Rank one: $(2, 0, 3)$	Fixed-kernel extension tests	5,995
Rank one: $(3, 2, 3)$	Recentering screen and fixed-kernel extension tests	5,995
Rank one: $(3, 0, 5)$	Recentering screen and binary projection filter	118

The repeated class lists in the rank-one rows are tested with different kernels. The table counts the full input classifications; the proofs below give the subsequent screens and search counts.

6.1 A forced support of size eight

Proposition 6.1 (Exact enumerator consequences)**.** *The hypothetical code is pure to distance six, contains a stabilizer word of weight eight, and satisfies*

$$A_6 + A_7 \leq 2. \tag{10}$$

Proof. The split-coset integer certificates for stabilizer weights one through five exclude all five possibilities without assuming purity. Hence $A_1 = \cdots = A_5 = 0$, which implies purity because the logical

distance is at least six. With these zeros imposed, the ordinary enumerator system plus $A_8 = 0$ has an exact infeasibility certificate with negative right-hand side $-3,932,160$.

For the last assertion, an integer combination of the ordinary MacWilliams and shadow relations gives $32768(A_6 + A_7) \le 65536$. Dividing by 32768 proves (10). Appendix A.4 gives the multipliers, row definitions, and sign conventions so that the identity can be checked directly. □

The following additive bound makes the forced support useful. It uses a parity property that is not visible in the first few nonnegative MacWilliams coefficients alone.

Lemma 6.2 (A nine-coordinate dimension bound)**.** *An additive code on nine sites with minimum distance at least six has binary dimension at most six.*

Proof. A code of larger dimension contains a dimension-seven subcode. Suppose $|C| = 128$, with nonzero weight counts A_6, A_7, A_8, A_9 summing to 127. The first three nonnegative dual MacWilliams coefficients give

$$A_7 + 2A_8 + 3A_9 \le 102, \qquad A_8 + 3A_9 \ge 75, \qquad 3(A_7 + A_8) - 4A_9 \ge 17.$$

The first two imply $A_7 \le 3A_9 - 48$, hence $A_9 \ge 16$, and also $A_7 + A_8 \le 27$. The third then gives $A_9 \le 16$. Equality forces $(A_6, A_7, A_8, A_9) = (84, 0, 27, 16)$.

On a binary additive space, weight parity is a quadratic form: its failure to be additive is the symplectic product in (8). For a quadratic form ε on a c-dimensional binary space U, define its Gauss sum by

$$G(\varepsilon) = \sum_{v \in U} (-1)^{\varepsilon(v)}.$$

For weight parity, this is the number of even-weight words minus the number of odd-weight words. If the polar form has rank 2ρ, then $G(\varepsilon)$ is zero or $\pm 2^{c-\rho}$. Indeed, squaring the sum and substituting $y = x + z$ gives

$$G(\varepsilon)^2 = 2^c \sum_{z \in \operatorname{rad}(U)} (-1)^{\varepsilon(z)}.$$

The restriction of ε to the radical is linear. The last sum is therefore zero if that restriction is nonzero, and otherwise equals $|\operatorname{rad}(U)| = 2^{c-2\rho}$. This proves the asserted possibilities. The forced distribution would instead give Gauss sum $128 - 2 \cdot 16 = 96$. Since $c = 7$ and $0 \le \rho \le 3$, the possible nonzero absolute values are $128, 64, 32, 16$, none of which is 96. This proves the bound. □

Choose a weight-eight stabilizer word p, put $T = \operatorname{supp} p$, and normalize $p = Z^8$. Let $R = S \cap V_T$, $r = \dim R$. The shortened centralizer has length nine and dimension $4 + r$, so Lemma 6.2 gives $1 \le r \le 2$.

If $r = 1$, then $R = \langle Z^8 \rangle$. If $r = 2$, choose $\zeta \in R \setminus \langle p \rangle$. Let α be the number of sites where ζ has an X-component. It is even by commutation. Since both ζ and $\zeta + p$ have weight at least six, $8 + \alpha \ge 12$, so $\alpha \in \{4, 6, 8\}$. Phase maps fixing Z remove Y on those sites. On the other $8 - \alpha$ sites, let β count the Z's of ζ. Then $\alpha + \beta \ge 6$ and $8 - \beta \ge 6$. Replacing ζ by $\zeta + p$ exchanges β with $8 - \alpha - \beta$. Consequently $\alpha = 4$ forces $\beta = 2$; at $\alpha = 6$ we may take $\beta = 0$ or one; and $\alpha = 8$ forces $\beta = 0$. Site permutations now give the following list.

$\dim R$	Generators of R	Nonzero local weights	$\dim C$
1	Z^8	8	5
2	$Z^8,\ XXXXZZII$	6,6,8	6
2	$Z^8,\ XXXXXXII$	6,8,8	6
2	$Z^8,\ XXXXXXZI$	7,7,8	6
2	$Z^8,\ X^8$	8,8,8	6

This list covers every possible local stabilizer, with $R = S \cap V_T$. The shortened-code classifications contain 32,865 dimension-five classes and 56 dimension-six classes. Section 7 gives their completeness arguments.

6.2 The four rank-two types

For the types with local weights $(6, 6, 8)$ and $(7, 7, 8)$, the local plane contains two words of weight six or seven. Equality therefore holds in (10). Every stabilizer word outside R has weight at least eight. For type $(6, 8, 8)$, the possible global profiles are $(A_6, A_7) = (1, 0), (1, 1), (2, 0)$. After removing the known local weight-six word, the allowed counts outside R are $(0, 0), (0, 1), (1, 0)$. These restrictions count distinct stabilizer words.

The adapted Z-only quotient in each rank-two type has dimension six. Every nonzero class has minimum weight at most four, since addition of Z^8 complements a Z-only word's support. The extension map M must be injective on this subspace at distance six. Classes of minimum weight one give the following necessary counting conditions: type $(6, 6, 8)$ requires four distinct labels of coset distance at least five, and the next two types require two. Of the 56 shortened-code classes, these tests leave 48, 52 and 52, respectively.

Proposition 6.3 (Mixed rank-two types)**.** *The three local types with weights* $(6, 6, 8)$*,* $(6, 8, 8)$*, and* $(7, 7, 8)$ *admit no extension to the hypothetical code.*

Computer-assisted proof. For each remaining shortened code and allowed pair (A_6, A_7), enumerate every eligible first syndrome image and every compatible independent tuple of six Z-only images. For each tuple, solve every complementary-image assignment, including all diagonal freedom, using $\Gamma = MJM^{\mathsf{T}}$ and all distance inequalities in the original coordinates. Completed maps must also have the prescribed numbers of weight-six and weight-seven words outside R. The equations and pruning conditions are given in Appendix B.

The complete records contain 641 first-image cases for type $(6, 6, 8)$, 650 for $(7, 7, 8)$, and 1,950 for the three profiles of $(6, 8, 8)$. Every case is exhausted with no accepted completion. For the first type, no complete assignment to the six Z-only images passes the necessary conditions. The physical six-site permutation reduction used in the latter two searches preserves every complete extension; its argument is in Appendix B.6. The counting conditions and the completed searches together cover all 56 shortened-code classes in each applicable profile. □

For the full plane $R = \langle Z^8, X^8 \rangle$, the local centralizer has additional structure. Every coset outside R in the local centralizer has four words, each of weight at least six, whose total weight is 24. All four weights are six. The full local centralizer $H = S^{\perp} \cap V_T$ is even and isotropic. Its quotient $K = H/R$ is an isotropic logical subspace, so $q \leq 3$. The identities of Proposition 3.1 give

$$h = 2q, \qquad \dim N = q.$$

The dimension-six shortened-code classification has hull distribution $h = 0, 2, 6$ with 15, 33 and 8 classes; no hull-four class occurs. Only $q = 0, 1, 3$ remain.

Proposition 6.4 (Full rank-two plane)**.** *The local type $R = \langle Z^8, X^8 \rangle$ admits no extension.*

Computer-assisted proof. For $q = 1$, a word complementary to R has two occurrences of each Pauli letter; site permutations give one local type H. For $q = 3$, every local H contains a dimension-four subcode containing R. The complete eight-site distance-six dimension-four classification contains three isotropic classes with a full-support plane. Testing all 256 normalizer cosets for each gives one dimension-five local H type containing R. In both cases R consists of the three nonzero full-support words, so physical equivalences preserve the plane as a whole.

For every shortened-code class, enumerate every q-space $N \leq \mathrm{rad}(P)$, up to verified physical suffix automorphisms. A space with more than two words of weight six or seven violates (10) and is excluded directly. The remaining cases have 63 N-orbits for $q = 1$, and 164 for $q = 3$. All identifications between the quotient radical and the local kernel radical are considered, modulo physical prefix equivalence. For $q = 3$, the induced physical group on K has order 21. Its action on the 168 identifications is free: an invertible identification fixed by an element forces that element to be the identity. There are therefore eight orbits, each of size 21. There are $63 + 164 \cdot 8 = 1375$ extension problems with fixed kernels and radical identifications, covering all 41 nonzero-hull shortened-code classes. The N-orbits are obtained by enumerating all three lines in each hull-two space and all 1,395 three-dimensional subspaces in each hull-six space. The recorded orbits form a disjoint partition of these sets; their sizes sum to three or 1,395 for each shortened code. The search exhausts all assignments to the Z-only basis images. For each of the 2,960 surviving tuples, exhaustive search finds no assignment to the complementary images.

For the 15 hull-zero shortened-code classes, the verification considers every admissible pair (A_6, A_7), including $A_6 = A_7 = 0$ and the cases with larger minimum stabilizer weight. These are covered by separate profile exclusions or by completed extension searches with 72,827 first-image cases. The verification program checks both sets of records and their coverage of the allowed pairs. Together these cover all 56 shortened-code classes and all allowed global profiles. □

6.3 The five rank-one kernel cases

It remains to exclude $R = \langle Z^8 \rangle$, with $a = 14$, $b = 13$. The shortened code has binary dimension five.

Lemma 6.5 (A three-dimensional kernel bound)**.** *The extension kernel has dimension $q \leq 3$.*

Proof. Proposition 3.3 gives $q \leq 4$. If $q = 4$, the local centralizer $H = S^{\perp} \cap V_T$ has dimension five and 32 words. The additive mean-weight bound is $\sum_{w \in H} \mathrm{wt}(w) \leq 8 \cdot 32 \cdot 3/4 = 192$. There are 30 words outside R, each of weight at least six, and Z^8 has weight eight. If O words have odd weight, this gives $\sum \mathrm{wt}(w) \geq 188 + O$, so $O \leq 4$.

Weight parity is a quadratic form on $\mathbb{F}_2^5$. A nonzero such form has at least eight ones: the Gauss-sum formula used in Lemma 6.2 bounds its positive nontrivial sum by 16, while a nonzero linear form has sum zero. Parity vanishes on H, making H isotropic by (8). But $H \cap S = R$, so H/R embeds as a four-dimensional isotropic subspace of the six-dimensional logical symplectic space. Its maximum isotropic dimension is three, a contradiction. □

Here $h = 2q - t - 1 \geq 0$ forces $q \geq 1$. The restricted rank t is even and satisfies $0 \leq t \leq q$; also $h \leq \dim C = 5$ and $q - t/2 \leq 3$. Together with $q \leq 3$, these conditions give exactly five possibilities. In each case $\dim N = q - 1$:

q	t	Hull dimension h	$\dim N$	Classes of C
1	0	1	0	26,752
2	2	1	1	26,752
2	0	3	1	5,995
3	2	3	2	5,995
3	0	5	2	118

The repeated shortened-code lists correspond to different kernels and must each be tested. Their union consists of all 32,865 classes.

6.4 Prescribing both kernels

Fix an exact $N \leq \mathrm{rad}(P)$ of the required dimension. The map τ descends to an injective isometry

$$P/N \longrightarrow K_E^{\perp}.$$

Its domain has dimension $14-q$ and radical dimension $g = q-t$. The radical maps isomorphically to $\mathrm{rad}(K)$. The complete local prefix classifications, with the full-support radical word p distinguished, have the following numbers of types:

(q,t)	$(1,0)$	$(2,2)$	$(2,0)$	$(3,2)$	$(3,0)$
$\#(H,p)$	4	2	6	5	7.

Here $H = S^{\perp} \cap V_T$ is the local centralizer, so $H/R = K$. A marked pair (H,p) records both this local space and the chosen weight-eight stabilizer word. An identification of radicals is a linear isomorphism between the radical of P/N and $\mathrm{rad}(K)$; it specifies where those commuting directions go. The marked local classifications are checked independently by direct normalized enumeration for $q = 1$, and by every ordered one-coordinate lift of the complete seven-site parent classifications for $q = 2, 3$. Distinguishing p prevents equating local spaces whose chosen weight-eight words are inequivalent.

For a fixed local H, every radical identification is considered. There are $|\mathrm{GL}(g,2)| = 1, 1, 6, 168$ possibilities for $g = 0, 1, 2, 3$, respectively. Physical automorphism orbits reduce these choices without dropping extensions. For each representative, the search assigns the seven independent Z-images and completes all seven complementary images. It enforces the identification, commutation, kernel equations, and physical distance constraints. Appendix B.5 proves that these equations force $\ker M = K$ and $\ker \tau = N$.

The isotropic $q = 3, h = 5$ case also admits a necessary binary projection filter without choosing a particular local H. For a basis c_1, c_2, c_3 of $\mathrm{rad}(P/N)$, tag each physical prefix site by its three X-evaluations. Every nonzero tag-functional must have even support of size at least four, since the corresponding kernel class has minimum weight at least six. All tag-count multisets satisfying this condition are enumerated. This is a relaxation of complete local kernel classification and is sufficient for that case's search.

Proposition 6.6 (Complete rank-one exclusion)**.** *None of the five rank-one kernel cases admits an extension.*

Computer-assisted proof. Every possible N is enumerated as a subspace of the radical of P, up to physical automorphisms of the shortened code. Inequality (10) excludes spaces with three words

of weight six or seven. Before taking orbits, the five rows of the preceding table have $1, 1, 7, 7, 155$ choices of N per shortened code, respectively: these are the Gaussian binomial counts for subspaces of the indicated dimensions. Orbit lists partition these explicit finite sets. A further screen may use the already excluded rank-two types: if a weight-eight word of N has another independent word of N supported inside its support, then recentering on that word gives a rank-two local stabilizer, covered by Propositions 6.3 and 6.4. All other N's proceed to search.

For $(q, t, h) = (3, 0, 5)$, 5,749 N-orbits are screened and the remaining 7,065 have completed searches over the Z-only images and all complementary images, with no extension found. This covers all 118 shortened-code classes. For $(3, 2, 3)$, 20,791 N-orbits are screened and 20,215 are searched against each of the five distinguished local types, giving 101,075 completed fixed-H, N cases. This covers all 5,995 shortened-code classes in that case.

The remaining cases $(1, 0, 1), (2, 2, 1), (2, 0, 3)$ cover 32,747 distinct shortened-code classes. For the last case all seven radical lines are included before quotienting by physical suffix symmetry. Every distinguished local type, radical identification, and eligible first syndrome image is included. The 2,419,796 completed search records cover these cases. For every assignment to the Z-only images that passes the necessary tests, exhaustive search excludes all assignments to the complementary images.

Some local types give identical necessary searches over the Z-only images. These enumerations are shared, but the complementary-image search is performed separately for each type. Appendix C specifies how this reuse and the different implementations are checked. □

6.5 Completion of the proof

Proposition 6.1 applies to every hypothetical code, including one initially assumed degenerate. Lemma 6.2 and the explicit local normal forms reduce it to the five types listed above. Their shortened-code classifications are complete. The rank-two types are excluded by Propositions 6.3 and 6.4; the rank-one type is exhausted by Lemma 6.5, the five kernel cases, and Proposition 6.6. This proves the second assertion of Theorem 1.1.

7 Classification and computational proof

7.1 Physical equivalence and graph certificates

The equivalence group on ℓ Pauli sites is $G_\ell = S_\ell \ltimes \mathrm{GL}(2, 2)^\ell$, of order $\ell! 6^\ell$. A local invertible map permutes the three nonzero letters, and over $\mathbb{F}_2$ every such permutation is symplectic. It preserves physical weight, commutation, hull dimension, and coset distance. One representative per orbit is enough. Two codes in the same orbit differ only by a relabeling of coordinates and Pauli letters, so an extension exists for one if and only if a correspondingly relabeled extension exists for the other. An orbit is one equivalence class, and $\mathrm{Aut}\, C$ is the subgroup fixing a code C. Orbit–stabilizer gives $|G_\ell|/|\mathrm{Aut}\, C|$ distinct labeled copies of that code. Labeled means that the coordinate positions and the three nonidentity Pauli letters at each coordinate are fixed.

One graph representation has a vertex for each codeword, three letter vertices per coordinate, and a coordinate vertex joined to its three letters. Each codeword vertex is joined to the letter it uses at each nonzero coordinate. Codeword, letter, and coordinate vertices have separate colors. A color-preserving graph isomorphism gives exactly a site permutation and local letter permutations

carrying the code to the other code. Conversely, every physical equivalence induces a graph isomorphism. The permutation on word vertices is forced by their incidences, so there is no additional graph-automorphism kernel. Canonical graph certificates and automorphism orders are computed with nauty [14]. Here a canonical graph certificate is a representation that is the same exactly for isomorphic colored graphs. It recognizes duplicate code classes; it is different from an infeasibility certificate for an enumerator system. The original census uses a second encoding with letter triangles rather than coordinate vertices. The two encodings are compared using the same code generators.

7.2 Completeness by extension and incidence

We use both shortening and puncturing. Shortening restricts to the words that are zero in a coordinate before deleting it; puncturing deletes the coordinate from every word.

For lengthening a distance-d code by one coordinate, let the new coordinate image have binary rank $\rho \in \{0, 1, 2\}$. Its zero-symbol shortening is a parent P whose dimension is lower by ρ. The extension is specified by a ρ-space in the old Pauli quotient modulo P, with every nonzero coset minimum at least $d - 1$. The new nonzero symbol adds one unit of weight. Rank one has three possible nonzero output letters; rank two has six output isomorphisms. These exhaust all children, because every child has a unique shortening at the last physical coordinate. The reason for counting labeled children as well as equivalence classes is completeness. Producing pairwise inequivalent codes gives a lower bound on the number of classes. A complete parent list lets us count every possible child before identifying equivalences. If the listed child orbits account for that entire count, none is missing.

Let $\mathcal{L}_\rho(P)$ be the set of ρ-dimensional subspaces $U \leq V_\ell/P$ whose nonzero cosets have minimum weight at least $d - 1$. Put

$$e_0(P) = 1, \qquad e_1(P) = 3|\mathcal{L}_1(P)|, \qquad e_2(P) = 6|\mathcal{L}_2(P)|.$$

The factors three and six count the injections $\beta : U \to \mathbb{F}_2^2$. Each pair (U, β) defines the child $\{(p + \widetilde{u}, \beta u) : p \in P, u \in U\}$, independently of the chosen representatives $\widetilde{u}$. Conversely, shortening a child at its last coordinate recovers P, and the remaining cosets recover (U, β). Each labeled child therefore occurs exactly once. For a fixed child dimension c, the parents in the following sum have dimension $c - \rho$, and the children have dimension c. Orbit–stabilizer counting gives

$$\sum_{\rho,P} \frac{\ell! 6^\ell}{|\operatorname{Aut} P|}\, e_\rho(P) = \sum_{D} \frac{(\ell+1)! 6^{\ell+1}}{|\operatorname{Aut} D|} \tag{11}$$

The sum over P uses one representative of every parent class of dimension $c-\rho$ and minimum distance at least d; the sum over D uses the listed child classes of dimension c and minimum distance at least d. The left-hand side counts all labeled extensions of the complete parent lists. The right-hand side counts the disjoint orbits of the listed children. Each child is checked individually for rank, distance, and distinct canonical certificate. Equality, together with complete parents, leaves no room for a missing orbit. Completeness follows inductively from the parent lists.

For the sixteen-qubit application, the lengthening induction starts with the zero code on four coordinates and retains every parent dimension that can reach the target. The base is complete because a nonzero word on four coordinates cannot have weight five. The four required classifications have the following masses $\sum_C |G_\ell|/|\operatorname{Aut} C|$:

ℓ	Binary dimension	Classes	Labeled mass
7	4	43	21,776,221,440
8	6	1,579	98,005,190,492,160
9	8	2,298	7,958,195,288,064,000
10	10	37	2,333,060,795,842,560

The same induction gives no codes in the three larger-dimension sets used in Lemma 5.2. The source release includes `verification/release16/census/check_cells.py`, which reconstructs these classifications, checks the mass identity at every step, and compares the resulting canonical certificates with the representatives used in the searches.

The seventeen-qubit classifications start from an elementary base. For a binary dimension-three code, each coordinate map $\mathbb{F}_2^3 \to \mathbb{F}_2^2$ is determined up to local output equivalence by its row space. There are fifteen row spaces of dimension at most two: one zero space, seven lines, and seven planes. Every multiset of seven such types is tested, giving $\binom{21}{7} = 116280$ possibilities. Exactly 2,171 meet the dimension and distance-five conditions; their physical graph certificates give 32 classes. This establishes the complete seven-site dimension-three classification directly.

Higher-dimensional supercodes supply another incidence check. Each dimension-$(c+1)$ binary code has $2^{c+1}-1$ hyperplanes. Its dimension-c parent extensions correspond to nonzero quotient cosets of minimum weight at least d. Counting parent/supercode incidences gives the following identity, where $e(P)$ is the number of those nonzero cosets:

$$\sum_P \frac{|G_\ell|}{|\operatorname{Aut} P|}\, e(P) = (2^{c+1}-1) \sum_D \frac{|G_\ell|}{|\operatorname{Aut} D|}. \tag{12}$$

The repository also checks the equality separately for each parent orbit by identifying every hyperplane of every listed child. This prevents incorrect per-class counts from canceling in a grand total.

Using (12), the 32 dimension-three parents prove the 43-class seven-site dimension-four list complete; those prove the one-class dimension-five list complete. The three parent dimensions then prove the 26,299-class eight-site dimension-five distance-five list complete through (11). Its labeled mass is $1,593,099,450,132,480$.

Finally, puncturing any nine-site dimension-five distance-six code preserves dimension and gives one of those eight-site parents: a nonzero puncture kernel would be a weight-one word. The new coordinate is a pair of binary linear forms on $\mathbb{F}_2^5$. All 1,024 ordered pairs are tested per parent. A lift is valid if and only if every parent word of weight five receives a nonzero new symbol. The unrestricted valid pairs number 1,791,031, and local output equivalence gives 299,175 canonical lifts. Their complete physical certificate set contains 32,865 classes, with hull counts $26752, 5995, 118$ at hull dimensions one, three, five.

The nine-site dimension-six classification is independently checked by (12) from those dimension-five classes. Each child has 63 hyperplanes. The incidence count is $4,332,624,025,927,680$; the 56 listed child orbits have total labeled mass $68,771,809,935,360$, exactly the former number divided by 63. All 3,528 hyperplanes are identified, and the equality is checked pointwise across all 32,865 parents. These checks establish completeness of the classifications used in the seventeen-qubit extension proof.

7.3 The local code classifications

The local centralizer codes used in Section 6 are obtained by puncturing and lifting. Puncturing an eight-coordinate code of minimum distance six preserves its binary dimension and gives a seven-

coordinate code of minimum distance at least five. For dimension four, the 43 parent classes above suffice. Testing all $2^8 = 256$ ordered pairs of linear forms per parent gives 861 valid lifts and 28 equivalence classes. Fifteen classes contain a word of weight eight; six of these are isotropic. Exactly three contain a full-support plane. Writing B for each of these three codes, we test all 256 cosets in $B^\perp/B$. In total, two cosets have minimum weight six, and the resulting dimension-five codes form one equivalence class. This supplies the local code needed in Proposition 6.4.

For the rank-one case, a code H of dimension $q+1$ is marked by a full-support word $p \in \mathrm{rad}(H)$. At $q = 1$, normalize $p = Z^8$ and test every possible second generator, modulo addition of p. At $q = 2, 3$, lift every seven-coordinate parent of dimension three or four and mark every full-support radical word. The complete parent lists have 32 and 43 classes. The marked physical graph certificates give, respectively, four types at $(q, t) = (1, 0)$, six and two at $(2, 0), (2, 2)$, and seven and five at $(3, 0), (3, 2)$. Marking p means giving its word vertex a separate color. An isomorphism must preserve the selected word. The repository includes a fresh coordinate-lift checker for the full-plane classifications; the complete marked-code check is identified in Appendix C.

7.4 Extension computations

For each shortened code, the search computes all coset minimum weights and enumerates the extension maps in the domain established by the preceding reductions. In the seventeen-qubit cases, it first assigns the images of a totally isotropic half of the local quotient and then solves for every complementary image. The Gram equation determines part of the latter assignment; all remaining entries, including diagonal entries, are searched. Appendix B gives the equations, pruning conditions, and symmetry arguments.

The computation is divided by shortened-code class, local kernel, radical identification, and first syndrome image. A case is excluded only when every part has been exhausted and no extension satisfies the required equations and weights. Interrupted computations are rerun to completion. The verification programs reconstruct this decomposition and check the completion records for every expected part.

7.5 Implementation checks

Known $[\![5, 1, 3]\!]$, $[\![16, 4, 5]\!]$, and $[\![16, 1, 6]\!]$ codes provide positive tests. Seventeen-coordinate $[\![17, 3, 3]\!]$ examples additionally test prescribed local kernels, shortened-code kernels, and singular Gram matrices. Their stabilizers and centralizers are constructed explicitly; ranks, commutation, and minimum distance are checked by enumeration.

The complementary-image algorithm is compared with direct enumeration on small examples, including assignments with nonzero diagonal entries. For the mixed rank-two cases, these comparisons cover 189,860 tuples in 1,804 fixed-tuple and distance-threshold instances. Separate C implementations are compared on common inputs. Tests with deliberately incomplete or inconsistent records check that these cannot be accepted as complete exclusions. The reproducibility appendix gives the precise scope of each verification command.

8 Conclusion

The two exclusions close the stabilizer-code distance intervals at $(n, k) = (16, 5)$ and $(17, 3)$. The proofs combine restrictions on a small supported stabilizer with complete classifications on the com-

plementary coordinates. Nonadditive quantum codes are outside this extension criterion, which uses a binary stabilizer subspace.

The remaining entries $[\![18, 4, 6]\!]$ and $[\![23, 1, 9]\!]$ in Grassl's tables [3] provide further targets. Lai's remaining candidate sets [16], Section VII-D, give concrete weight-enumerator inputs. Applying the method would require a forced support with a classifiable complement, followed by kernel and coset-weight tests for its possible local stabilizers.

The hypotheses of the extension criteria determine which search is applicable. Theorem 2.1 requires all short stabilizer words to lie in the chosen local stabilizer. The version for arbitrary degeneracy uses a relative-distance function that depends on the extension map. The balanced-plane bound assumes that the entire local stabilizer is the specified plane.

Benchmark data and programs are retained in `results/benchmarks/` and `source/benchmarks/`.

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A Exact weight-enumerator certificates

This appendix specifies the necessary linear systems used for the low-weight reductions. The certificate files contain their integer multipliers and all case assumptions. The checker rebuilds the columns from physical Pauli cosets.

A.1 Split transforms and the affine shadow

The ordinary weight enumerator records only total weight. A split enumerator also records which coordinate blocks contain the nonidentity letters. This is needed when a specified stabilizer word or subgroup occupies one block. The formulas below are the MacWilliams and shadow identities applied separately to each block.

Partition the physical coordinates into blocks of sizes $n_1, \dots, n_s$. Let $A^{\mathrm{split}}_{\mathbf{i}}$ count stabilizer words with split weight $\mathbf{i} = (i_1, \dots, i_s)$. Put $B = |S| = 2^{n-k}$, and

$$\mathcal{K}^{\ell}_{j}(w) = \sum_{u=\max(0,j-\ell+w)}^{\min(j,w)} (-1)^u 3^{j-u} \binom{w}{u} \binom{\ell - w}{j - u}.$$

Character orthogonality gives the centralizer count

$$|\{v \in S^{\perp} : \mathrm{splitwt}(v) = \mathbf{j}\}| = B^{-1}\mathcal{T}_{\mathbf{j}}, \quad \mathcal{T}_{\mathbf{j}} = \sum_{\mathbf{i}} \left(\prod_{\ell} \mathcal{K}^{n_\ell}_{j_\ell}(i_\ell) \right) A^{\mathrm{split}}_{\mathbf{i}}.$$

The character identity behind this formula is

$$\mathbf{1}_{v \in S^{\perp}} = B^{-1} \sum_{s \in S} (-1)^{\langle v, s \rangle}.$$

The sum is B when every word of S commutes with v, and zero otherwise. Summing it over words of a fixed split weight produces the products of Krawtchouk polynomials above. This explains why the

transformed coefficients count actual centralizer words. On an isotropic stabilizer the parity function is linear, by (8). The affine set

$$\mathcal{A} = \{v : \langle v, s\rangle = \mathrm{wt}(s) \bmod 2 \text{ for every } s \in S\}$$

is a nonempty coset of $S^{\perp}$, since every linear functional on S extends to the ambient space. Character orthogonality gives its split weight counts as $B^{-1}\widehat{\mathcal{T}}_{\mathbf{j}}$, where

$$\widehat{\mathcal{T}}_{\mathbf{j}} = \sum_{\mathbf{i}} (-1)^{\sum i_\ell} \left(\prod_{\ell} \mathcal{K}_{j_\ell}^{n_\ell}(i_\ell) \right) A_{\mathbf{i}}^{\mathrm{split}}.$$

Both transforms are nonnegative. Centralizer inclusion and distance give

$$\mathcal{T}_{\mathbf{j}} \geq B A_{\mathbf{j}}^{\mathrm{split}}, \qquad \mathcal{T}_{\mathbf{j}} = B A_{\mathbf{j}}^{\mathrm{split}} \quad (1 \leq \textstyle\sum j_\ell < d).$$

These statements hold without assuming purity.

A.2 Coset-pattern columns

If a specified isotropic subgroup R_i lies on block i, enumerate every coset of R_i in $R_i^{\perp}$. Its pattern $p = (p_0, \ldots, p_{n_i})$ counts words at each physical weight. Let $y \geq 0$ count stabilizer cosets aggregated by their block patterns and suffix weight. Then a split count is the corresponding linear combination of products of the p_j's. With no prescribed subgroup a block may instead use the split counts themselves as variables. Every actual stabilizer containing the prescribed subgroups gives such nonnegative variables, since it is closed under their addition. The model need not reconstruct a linear code from an arbitrary feasible point; only necessity is used.

The row families are:

Row	Definition
Unit	$A_{\mathbf{0}}^{\mathrm{split}} = 1$.
Total	$\sum A_{\mathbf{i}}^{\mathrm{split}} = B$.
Low distance	$\mathcal{T}_{\mathbf{j}} - B A_{\mathbf{j}}^{\mathrm{split}} = 0$, $1 \leq \sum j_\ell < d$.
Inclusion	$B A_{\mathbf{j}}^{\mathrm{split}} - \mathcal{T}_{\mathbf{j}} \leq 0$.
Centralizer	$-\mathcal{T}_{\mathbf{j}} \leq 0$.
Shadow	$-\widehat{\mathcal{T}}_{\mathbf{j}} \leq 0$.
Weight assumptions	$\sum_{\sum i_\ell = w} A_{\mathbf{i}}^{\mathrm{split}} = A_w$, or its explicitly assumed lower bound.
Coset relations	A specified linear relation valid separately on every physical pattern.

The checker verifies each coset relation against the regenerated patterns. No property of a later shortened code is inserted into these reductions. The variables may be allowed to take nonnegative real values. This enlarges the set of possible enumerators, so infeasibility of the resulting linear system still excludes every actual code. Integrality is used separately in the explicitly stated parity and counting steps.

A.3 The sign convention

Write the system as $Ez = b$, $Gz \leq g$, $z \geq 0$. A certificate consists of arbitrary integer equality multipliers ν and nonnegative integer inequality multipliers λ with

$$c = \nu E + \lambda G \geq 0, \qquad \nu b + \lambda g < 0.$$

The condition $c \geq 0$ is checked coordinate by coordinate: since all variables in z are nonnegative, it makes $cz \geq 0$. Equality rows may be multiplied by either sign; inequality rows need nonnegative multipliers so their directions are preserved. For a feasible z, one would have $0 \leq cz \leq \nu b + \lambda g < 0$, a contradiction. The checker tests every coefficient column and the negative right-hand side with exact integers. The multipliers may be found by numerical optimization; the certificate is checked using integer arithmetic alone.

A.4 The seventeen-qubit ordinary identity

For the bound on $A_6 + A_7$ in Proposition 6.1, write $\mathcal{K}_j(w) = \mathcal{K}_j^{17}(w)$ and let $B = 2^{14}$. The low-weight rows are $\sum_w \mathcal{K}_j(w)A_w - BA_j = 0$ for $1 \leq j \leq 5$, and let the shadow rows be $-\sum_w (-1)^w \mathcal{K}_j(w)A_w \leq 0$. Combine the unit row $A_0 = 1$ with multiplier $-1,835,008$, the total row with multiplier 116, the five low-weight rows with $(60, 25, 11, 3, 1)$, the five pure-zero rows with $(-851968, -278528, -507904, -147456, -180224)$, and the shadow rows $j = 0, \ldots, 4$ with $(88, 32, 18, 4, 2)$. The result is $32768(A_6 + A_7) \leq 65536$. Every coefficient identity is checked independently in the repository.

A.5 Main certificate inventory

The repository groups certificates by target parameters. "Columns" means the number of nonnegative variables in the expanded system.

Code	Case	Columns	Negative right-hand side
$[\![16, 5, 5]\!]$	Weight 1	16	$-11,520$
	Weight 2	45	$-374,784$
	Weight 3	42	$-5,662,720$
	Weight 4, $A_4 \geq 2$	78	$-7,294,976$
	Weight 5, $A_1 = \cdots = A_4 = 0$	72	$-8,192$
	Minimum stabilizer weight at least 10	17	$-61,440$
$[\![17, 3, 6]\!]$	Weight 1	17	$-786,432$
	Weight 2	48	$-5,324,800$
	Weight 3	45	$-442,368$
	Weight 4	84	$-81,920$
	Weight 5	78	$-114,688$
	Pure, $A_8 = 0$	18	$-3,932,160$

The extra seventeen-qubit certificate excluding minimum stabilizer weight at least nine is also replayed, but is not needed for the forced-weight-eight route once the $A_8 = 0$ certificate has been used. The thirteen simple local weight-count certificates express the local plane's necessary weight-six or weight-seven counts; their inequalities are already visible in Section 6.

The fourteen-qubit bundle contains the overlap and rank-three certificates of Proposition 4.1, with its original standalone checker. The ordinary fourteen-qubit identities and the sixteen-qubit parity congruence are also replayed separately.

B The finite extension searches

B.1 Enumeration of the extension maps

For each classified shortened code, the following procedure is applied. The local quotient has a fixed physical symplectic basis of size 2μ. The search first chooses the images of one isotropic half of this basis, then chooses the images of its symplectic partners. In matrix notation these two groups of columns are X and Y. A partial assignment is discarded only when it already violates a necessary condition; every assignment that survives is extended through all remaining allowed values.

1. Enumerate the permitted local kernels K, shortened-code kernels N, and radical identifications, retaining representatives under the physical groups specified below.
2. Assign the μ images X one at a time. Test all newly determined linear combinations, the kernel evaluations, and the necessary partial-rank and weight-count conditions.
3. For a complete X, transform the Gram matrix as in (13). Reject it if the lower-right block is nonzero.
4. Enumerate the remaining free entries of U, including its diagonal. Test the kernel equations and every complementary mask against its full distance domain $\mathcal{D}_x$.
5. Reconstruct each completed map in the original physical coordinates and test the required stabilizer weight counts. Save any survivor.

The sixteen-qubit relaxation omits the Gram and prescribed-kernel tests and assigns all basis images using only the distance inequalities. Every genuine extension occurs in the classified domain, passes the necessary partial tests, and is retained by the physical symmetry reductions. If no completed assignment passes the final tests, no genuine extension exists. A survivor of the sixteen-qubit relaxation would require a separate commutation check.

B.2 Coset distances from physical words

Using a parity-check basis of C, compute the syndrome of each word $w \in V_f$ and minimize $\mathrm{wt}(w)$ for each syndrome. Equivalently, construct the Cayley graph of W/C whose edges add one nonzero Pauli at one site, and use breadth-first search from zero. A path of length u gives a physical word of weight at most u, while a word of weight v gives a path of length v. Therefore the two procedures compute the same exact coset minimum. Rank checks verify that the syndrome kernel is C.

B.3 Completing a partial extension

Write $\dim E = 2\mu$, choose a symplectic basis, and write $M = [X\ Y]$, where X contains images of classes represented using only the letters I and Z. We call these the Z-only classes. Every nonzero Z-only class has minimum weight below the target distance in the searches that use this normal form. Its image cannot vanish, so X has rank μ. Complete its columns to an invertible syndrome basis B_X, and put

$$X' = B_X^{-1}X = \begin{bmatrix} I_\mu \\ 0 \end{bmatrix}, \quad Y' = B_X^{-1}Y = \begin{bmatrix} U \\ L \end{bmatrix}, \quad \Gamma' = B_X^{-1}\Gamma B_X^{-\mathsf{T}}.$$

The compatibility equation is

$$\Gamma' = \begin{bmatrix} U + U^{\mathsf{T}} & L^{\mathsf{T}} \\ L & 0 \end{bmatrix}. \tag{13}$$

The lower-right block must vanish; L is fixed; and $U + U^{\mathsf{T}}$ fixes each off-diagonal pair sum while leaving every diagonal free. For $\mu = 6$, there are 21 binary free entries before distance and kernel constraints.

For each nonzero complementary mask x, form the domain

$$\mathcal{D}_x = \{u \in \mathbb{F}_2^{\mu} : \delta(B_X(z + u, Lx)) + c_R(z, x) \geq d \text{ for every } z \in \mathbb{F}_2^{\mu}\}.$$

Here x is a binary coefficient vector selecting a sum of the complementary basis vectors, while z selects a sum of the first basis vectors. The matrix L is already fixed. For a proposed value $Ux = u$, the displayed test checks all words with those complementary coefficients, by letting z run over $\mathbb{F}_2^{\mu}$. The remaining exact constraints are $Ux \in \mathcal{D}_x$ for all $x \neq 0$, together with (13) and any prescribed kernel equations. A finite traversal of all allowed entries of U therefore enumerates every completion. Consistency tests on pairs of assignments prune the domains. The final check uses every mask, including those of size greater than two.

For a fixed kernel vector with coordinates (z, x), the additional equation is $Xz + Yx = 0$; in this normal form it reads $Ux = z$, $Lx = 0$. The implementation retains the resulting affine freedom and exhausts all completions for each retained assignment to X.

B.4 Partial rank and exact profile tests

After assigning j Z-only images, let $\mathcal{U}_j$ be their common annihilator in the coefficient space of P. Restricting $\Gamma = XY^{\mathsf{T}} + YX^{\mathsf{T}}$ to $\mathcal{U}_j$ makes the assigned terms vanish. Only $\mu - j$ pairs remain, so the restricted rank is at most $2(\mu - j)$: each remaining pair contributes a matrix $uv^{\mathsf{T}} + vu^{\mathsf{T}}$ of rank at most two. This is a necessary partial-depth test, with a zero annihilator block at full depth.

We also use the weight distribution of the reconstructed stabilizer. For a coefficient vector a, write $v(a) = \sum a_i v_i$ and $e(a) = JM^{\mathsf{T}}a = (Y^{\mathsf{T}}a, X^{\mathsf{T}}a)$. Its words are

$$(\widetilde{e}(a) + r, v(a)), \qquad a \in \mathbb{F}_2^{b},\ r \in R. \tag{14}$$

All pairs (a, r) give distinct physical words, since the suffix basis is independent. The words with $a = 0$ are R. This remains true when Γ is singular.

At full X, for $a \in \mathrm{Ann}(X)$, compatibility gives $\Gamma a = X(Y^{\mathsf{T}}a)$. Independence of X determines $Y^{\mathsf{T}}a$ uniquely when it exists. Every translate with such a coefficient can then be counted exactly. The estimates below are used only for the other coefficients, so each word is counted once. At completed M, all words in (14) are counted exactly and the required profile equality is checked. No inversion of a singular Γ is used.

For the remaining coefficients, we obtain lower bounds on the number of weight-six and weight-seven words as follows. For a fixed nonzero coefficient a, let $\mathcal{P}_a$ contain every prefix representative that any completion of the current partial map could assign to $v(a)$. A nonempty superset is sufficient; if there are no possible prefixes, the partial assignment has no completion. Define

$$U_a = \mathrm{wt}(v(a)) + \max_{v \in \mathcal{P}_a} \min_{r \in R} \mathrm{wt}(v + r).$$

The order of the maximum and minimum is important. For each possible prefix, first take its least weight modulo R; then take the largest value any completion could achieve. Even this best possible completion cannot make the corresponding coset heavier than U_a. For every completion, the minimum weight in the corresponding R-coset is at most U_a. In the pure distance-six setting, $U_a \leq 7$

therefore guarantees at least one word of weight six or seven, and $U_a \leq 6$ guarantees a weight-six word. Distinct coefficients give disjoint cosets by (14), so these guaranteed counts may be added. If they exceed the allowed counts outside R, no completion exists. The same argument applies to bounds on individual translates. Because the maximization uses a superset of the possible prefixes, it can weaken the rejection test but cannot remove a valid completion. Exact profile equalities are tested only after the whole map is assigned.

B.5 Exact prescribed kernels

For fixed $N \subseteq \mathrm{rad}(P)$, work in $\bar{P} = P/N$. Let its alternating matrix be $\bar{\Gamma}$. A genuine extension gives an injective isometry $\tau : \bar{P} \to K^{\perp}$. Let $g = q - t$; fix every isomorphism from $\mathrm{rad}(\bar{P})$ to $\mathrm{rad}(K)$. Relative to chosen bases, this specifies evaluations of the Z-only image columns against every radical coefficient. The coordinate labels given by these g radical evaluations span $\mathbb{F}_2^g$. A site permutation makes the first g labels independent. The same permutation is applied to the local quotient basis and its table of coset minimum weights.

Subtract suitable combinations of those first g image columns to obtain vectors $T_i \in \mathrm{im}\,\bar{\Gamma}$, and make the same subtractions from the physical prefix vectors. When $t = 2$, project the latter off a fixed symplectic pair spanning the nondegenerate part of K. Call the resulting vectors e_i. For any solution c_i of $\bar{\Gamma} c_i = T_i$, compatibility requires

$$c_i^{\mathsf{T}} T_j = \langle e_i, e_j \rangle.$$

Changing c_i by a radical vector does not alter the left-hand side. These equations, all radical evaluations, and all Z-only weight inequalities are necessary conditions during the search over X. Every retained assignment is then tested by the complete search over Y.

The final completion enforces $M(K) = 0$ and the full Gram equation. The prescribed radical identification also fixes the values of X^{T} on $\mathrm{rad}(P)$. Its kernel there is N: on $\mathrm{rad}(P)/N$ it is the identification with $\mathrm{rad}(K)$, followed by projection onto the second half of the local coordinates. That projection is injective on K, because a nonzero element in its kernel would be a Z-only class of weight less than d. The following argument shows why these conditions give the prescribed kernels exactly.

Identify P with its coefficient space $\mathbb{F}_2^b$, so its radical is $\ker \Gamma$, and write $\tau = JM^{\mathsf{T}}$. If $\tau c = 0$, the Gram equation gives $\Gamma c = 0$; also $X^{\mathsf{T}} c = 0$. The prescribed radical evaluations therefore give $c \in N$. Conversely, if $c \in N$, then $\Gamma c = 0$ and $X^{\mathsf{T}} c = 0$. Expanding the Gram equation,

$$0 = \Gamma c = X(Y^{\mathsf{T}} c) + Y(X^{\mathsf{T}} c) = X(Y^{\mathsf{T}} c).$$

The columns of X are independent, so $Y^{\mathsf{T}} c = 0$. $\tau c = 0$, proving $\ker \tau = N$. Finally,

$$\mathrm{rank}\, M = \mathrm{rank}\, \tau = b - \dim N = 2\mu - q, \qquad \dim \ker M = q.$$

Since $K \subseteq \ker M$ and $\dim K = q$, we obtain $\ker M = K$. Requiring containment of the prescribed subspaces therefore suffices here because the radical conditions also fix the dimensions; larger kernels cannot pass these tests.

B.6 Physical symmetry and joint normalization

The symmetry reductions use site permutations and local permutations of the nonzero Pauli letters. These preserve both the symplectic form and the minimum weights in the original coordinates.

A physical suffix automorphism acts on coefficient vectors by G_{coef}: $g(v(a)) = v(G_{\mathrm{coef}}a)$. Its syndrome action is $G_{\mathrm{syn}} = G_{\mathrm{coef}}^{-\mathsf{T}}$, since $\sigma(gw) = G_{\mathrm{coef}}^{-\mathsf{T}}\sigma(w)$. $M \mapsto G_{\mathrm{syn}}M$, and $G_{\mathrm{syn}}\Gamma G_{\mathrm{syn}}^{\mathsf{T}} = \Gamma$. This contragredient action is essential when Γ is singular. Every generator is checked on physical words, coset distances, and the corresponding form.

Outer prefix automorphisms may reduce radical identifications if they preserve the local space and distinguished local stabilizer. Inside a fixed identification, only actions fixing its radical classes are used. Suffix actions likewise preserve the specified C, N and radical identification. These restrictions ensure that both actions preserve the set of extensions with the prescribed kernels and identification.

Choose a lexicographically least *whole* Z-only tuple in a joint orbit of those prefix and suffix actions. It is lexicographically minimal under the prefix action, and its first label is least in its suffix orbit. Otherwise one of the actions would reduce the same whole tuple. Imposing both conditions therefore retains a representative of every joint orbit, even when the two actions do not commute. No uniqueness of a joint canonical form is required.

For the mixed six-site permutation reduction, the fixed first image is unchanged and the other images are differences from an anchor among six points. Physical permutations reorder and reanchor these points. A lexicographically least full tuple is increasing and no translated anchor tuple is smaller. The search assigns points in increasing order, so its current tuple is a fixed initial segment of every completion of that branch. Under a change of anchor, adding future points can only decrease the initial entries of the sorted translated list. If the already assigned points give a smaller initial segment after translation, every completion has a physically equivalent tuple smaller than itself. Such a branch cannot contain the least representative and can be discarded. The same physical action is transported to the complementary basis, including its possible shear. Optional relaxed complementary-domain filters that are not invariant under reanchoring are disabled in that search; the complete Y-test is retained.

The full-plane eight-site reduction uses the corresponding physical eight-point action and verifies it on the full quotient basis. All first-label orbit lists explicitly record every eligible label and its orbit membership. Each list is divided into consecutive blocks of at most 64 labels. Searching every block covers every eligible first image.

C Computational data and reproducibility

Data availability. The manuscript is archived at doi:10.5281/zenodo.23874369; the public repository is `https://github.com/brandonlign/quantum`. It contains the paper, exact certificates and standalone checks in `verification/`, byte-preserved research programs and inputs in `source/`, and saved reports in `results/recorded_checks/`. The evidence archive in the same Zenodo release (doi:10.5281/zenodo.23904905) supplies the search-record database `source/evidence/search_records.sqlite`, its index description `source/evidence/record_store_manifest.json`, and the classified inputs and aggregation files required by the full saved-evidence verifier. Extracting both archives into the same directory reconstructs the expected layout. The database contains original compressed record payloads in an indexed container; `source/record_store.py` reads their original bytes, and `source/materialize_records.py` extracts them to the original relative directory layout. The reproduction guide `REPRODUCING.md` gives the commands, dependencies, expected outputs, and scope of each verification tier.

C.1 Verification programs and their scope

`verification/replay_certificates.py` rebuilds the linear systems of Appendix A, checks 37 integer certificates, and checks four ordinary identities. Fourteen-qubit data and its standalone checker are in `verification/certificates/14/` and `verification/check_certificates14.py`. `verification/check_constructions.py` enumerates both matching constructions. The local classification and full-plane orbit checks are `verification/check_local_classification.py` and `verification/check_full_plane_orbits.py`. The sixteen-qubit census entry point is `verification/release16/census/check_cells.py`; its search programs are in `verification/release16/search/`. `SHA256SUMS` identifies the tracked files, excluding itself.

The original saved-evidence program is `source/quantum-extension/verification/verify_evidence.py`, with the storage entry point `source/verify.py`. Using its original inputs and records, it checks the classification completeness arguments in Section 7, the local kernel classifications, and the expected completed search parts. It recomputes the saved aggregations except for the three kernel cases $(q,t,h) = (1,0,1),(2,2,1),(2,0,3)$. Their saved aggregation covers 32,747 shortened-code classes and 2,419,796 search records. The default command checks full class coverage and source hashes and repeats the aggregation for eight complete classes containing 1,059 records. Its `--deep` option repeats that full aggregation. The final complete aggregation was performed on 16 October 2026. The original reports record which checks were repeated and which saved results were checked by their hashes. A separate integrity program, `source/verify_all_record_hashes.py`, compares each record from those three kernel cases with its saved SHA-256 digest and checks its completed-status fields.

C.2 Shared enumerations and implementation checks

Some rank-one $q = 1$ local types give identical necessary problems for the Z-only images. Their enumeration is shared after checking that the serialized inputs, group actions, program version, and options agree. Each target type retains its own kernel equations and exhaustive search over complementary images. The verifier checks the hashes of the shared complete enumerations before and after reuse.

The Mac and Linux implementations are assigned disjoint sets of code classes. The saved-evidence program checks this assignment, source and binary hashes, and completion of every expected search part. It rejects unrecognized versions, conflicting records, unfinished searches, and unresolved candidates. The implementations enumerate the same domain with different pruning tests. Their recorded reports are retained in `results/recorded_checks/`.

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